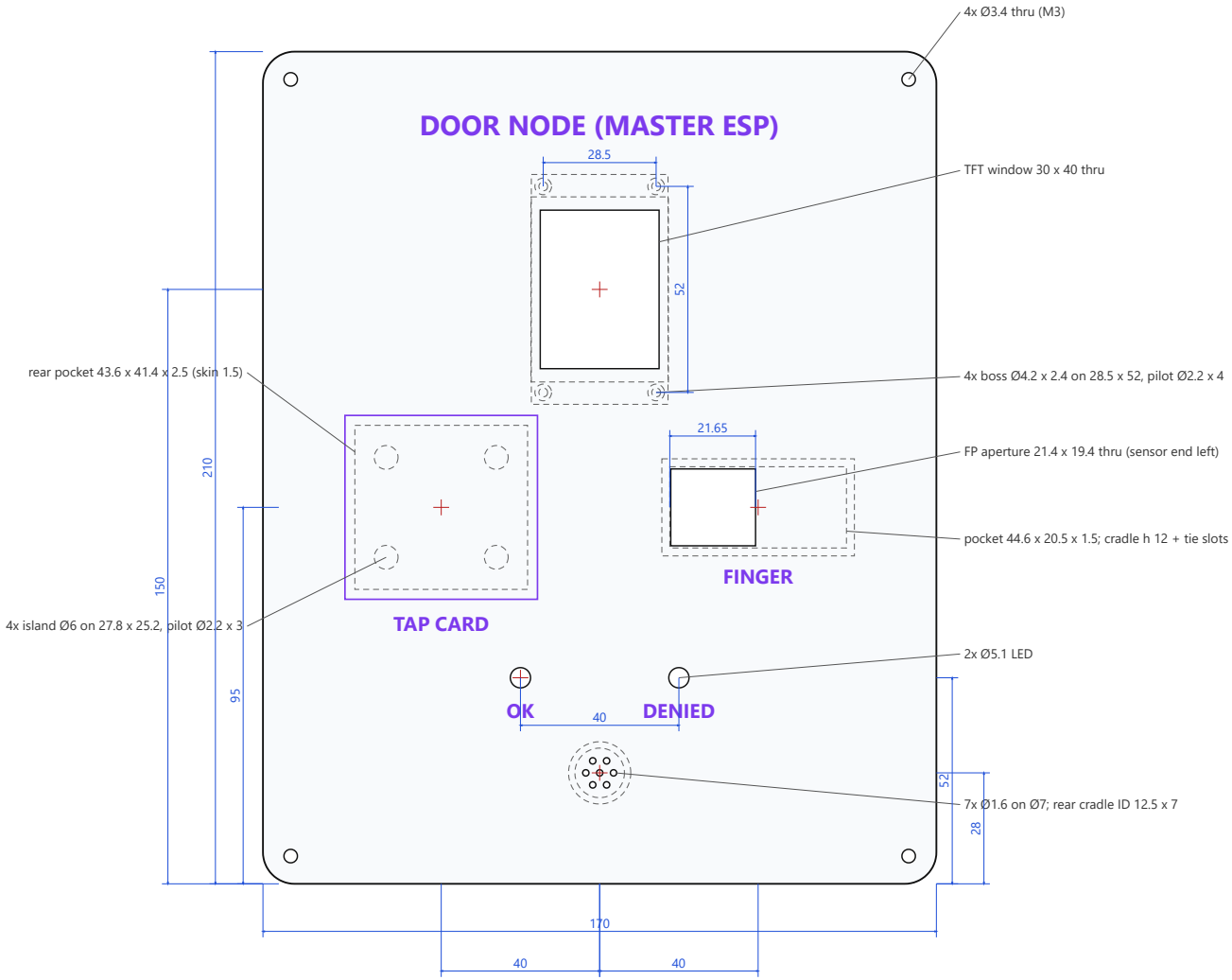
Front panel - front view

seen from the corridor; dashed = rear features (pockets, bosses, cradles); purple = engraved

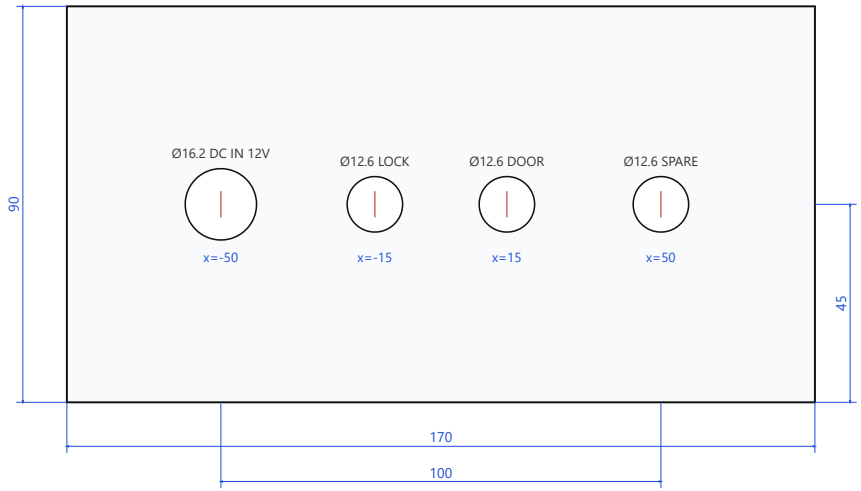


ALIoT Smart Lab - MASTER_LAB01 door node enclosure source: tools/cad/door_enclosure/params.json + Build.cs (SOLIDWORKS 2025)	units mm general tol. ±0.3 (FDM)	scale ≈1:1.8 2026-10-20	sheet 1 / 4 status: FIT-TEST RELEASE

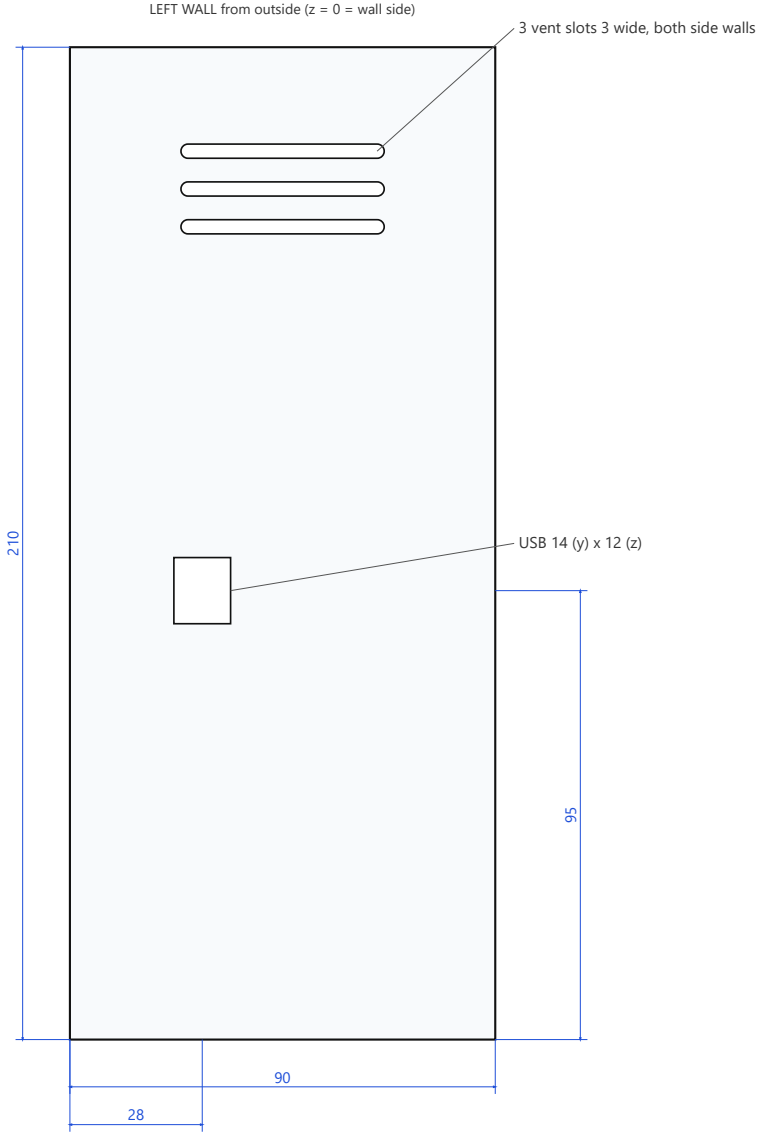
Body - bottom wall and left wall

cable entries (bottom) and ESP32 USB access + vents (left)

BOTTOM WALL from below (back face at the top edge = wall side, z = 0)



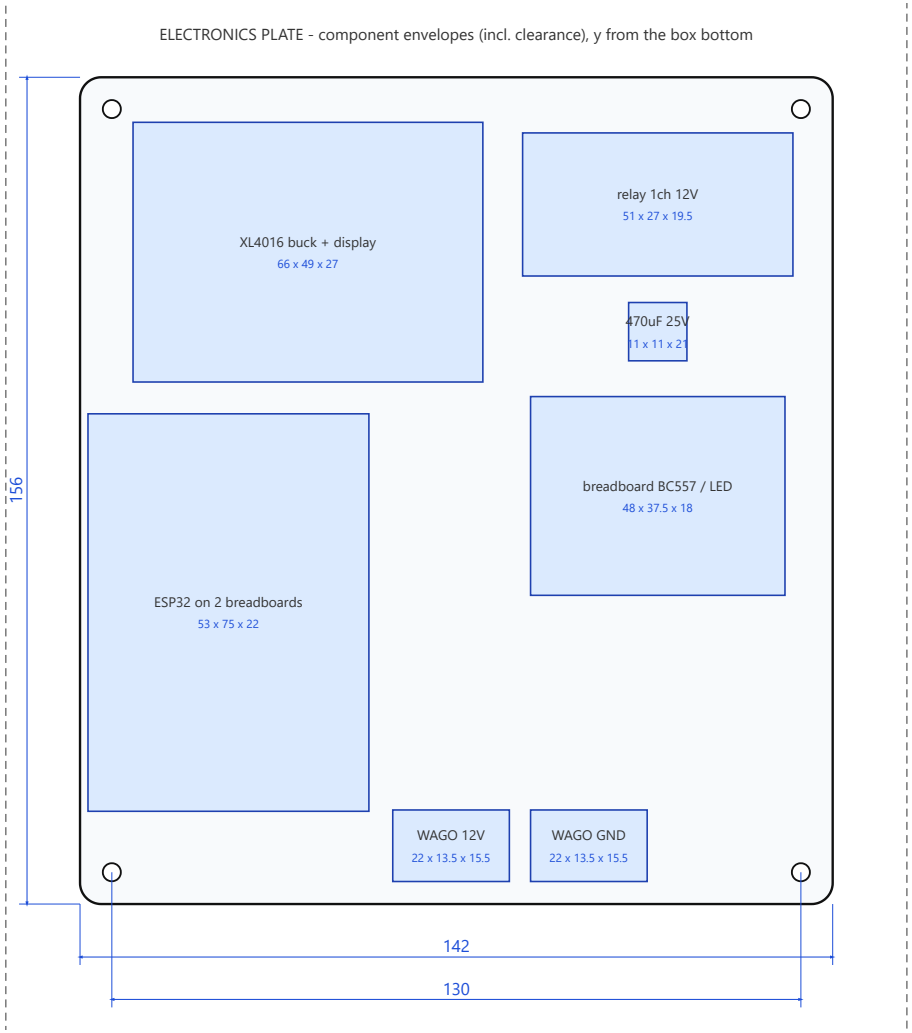
LEFT WALL from outside (z = 0 = wall side)



ALIoT Smart Lab - MASTER_LAB01 door node enclosure source: tools/cad/door_enclosure/params.json + Build.cs (SOLIDWORKS 2025)	units mm general tol. ±0.3 (FDM)	scale ≈1:1.6 2026-10-20	sheet 2 / 4 status: FIT-TEST RELEASE

Electronics plate - layout

component envelopes as used in the SOLIDWORKS interference check



ALIoT Smart Lab - MASTER_LAB01 door node enclosure source: tools/cad/door_enclosure/params.json + Build.cs (SOLIDWORKS 2025)	units mm general tol. ±0.3 (FDM)	scale ≈1:1.3 2026-10-20	sheet 3 / 4 status: FIT-TEST RELEASE

Component envelopes

every box in the virtual assembly, with where its size comes from

Envelope	w x h x d (incl. clearance)	centre x, y; base z	confidence	source
XL4016 buck + display	66 x 49 x 27	-28, 163, z 13 up	VARIANT (Most listing ~64 x 47 x 26) +1	mostelectronic.com XL4016 8A display
relay 1ch 12V	51 x 27 x 19.5	38, 172, z 13 up	REFERENCE 50 x 26 x 18.5 +1	generic SRD-12VDC module
470uF 25V	11 x 11 x 21	38, 148, z 13 up	REFERENCE D10 x 20 radial	typical 470uF 25V
ESP32 on 2 breadboards	53 x 75 x 22	-43, 95, z 13 up	VARIANT ESP32 51.5 x 28.5; REFERENCE SYB-170 47 x 36.5 x 10 (x2)	Most ESP32 30-pin; Most ME-29142
breadboard BC557 / LED	48 x 37.5 x 18	38, 117, z 13 up	REFERENCE SYB-170 + parts 8 mm	Most ME-29142
WAGO 12V	22 x 13.5 x 15.5	-1, 51, z 13 up	VARIANT PCT-212 21 x 12.5 x 14.5 +1	Most WAGO PCT-212
WAGO GND	22 x 13.5 x 15.5	25, 51, z 13 up	VARIANT PCT-212 +1	Most WAGO PCT-212
TFT glass	34.9 x 46.9 x 2.4	0, 150, z 90 down	DATASHEET 34.7 x 46.7 x 2.4 +0.2	QDTFT1801
TFT PCB + header plugs	34.5 x 58 x 15.6	0, 150, z 87.6 down	VARIANT PCB 1.6 + plugs 14	LCDWIKI MSP1803
PN532 + header plugs	43 x 41 x 17.6	-40, 95, z 90 down	VARIANT 42.7 x 40.4; plugs 16 REFERENCE	Ekostra EK506 / Elechouse
R307 body + connector	44.6 x 20.5 x 26.7	40, 95, z 91.5 down	DATASHEET 44.1 x 20 x 23.5 + 3.2 connector	GROW R307 manual
LED green + wires	6 x 6 x 16	-20, 52, z 90 down	REFERENCE flange 5.9	Kingbright L-53
LED red + wires	6 x 6 x 16	20, 52, z 90 down	REFERENCE flange 5.9	Kingbright L-53
active buzzer D12	12.4 x 12.4 x 21	0, 28, z 90 down	DATASHEET TMB12A05 (cylinder D12.4) 12 x 9.5 + pins 5.5 + wires	Huaneng TMB12A05 / Most ME-32997

ALIoT Smart Lab - MASTER_LAB01 door node enclosure source: tools/cad/door_enclosure/params.json + Build.cs (SOLIDWORKS 2025)	units mm general tol. ±0.3 (FDM)	scale - 2026-10-20	sheet 4 / 4 status: FIT-TEST RELEASE